



Course Topic: Couple Model (Climate Forecast system), Ensemble prediction system, multi-model ensemble technique,
Extended range forecasting.

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INDIA METEOROLOGICAL DEPARTMENT**



Operational Forecasting Perspectives

1. **Dynamical Forecast**
2. **Statistical/Machine Learning Forecast**

1. **Tropical Forecast**
2. **Extratropical Forecast**

1. **Weather System Forecast**
2. **Climate Forecast**

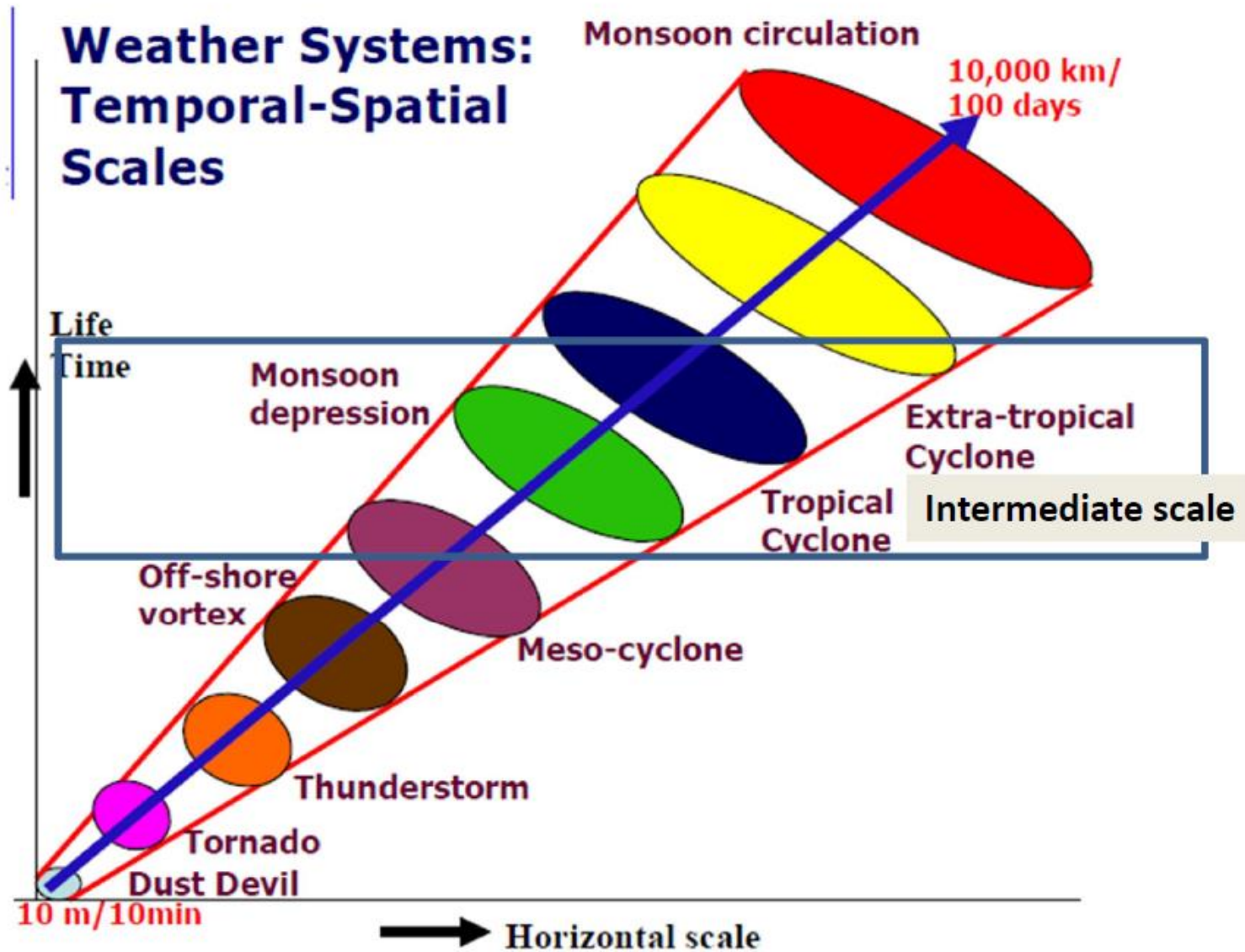
1. **Spatial Scale of Forecast**
2. **Temporal Lead time of Forecast**

1. **Deterministic Forecast**
2. **Probabilistic Forecast**

**Sector
Specific
Impact**



Different systems and their Scales



Predictability

The predictability of a weather system which has source in the knowledge of the initial conditions and which is limited by the flow itself is called **Predictability of the First Kind**.

Baroclinic Instability, Barotropic Instability,
Interactions of eddies



Predictability of a dynamical system: A Fundamental Question in Predictability

A Dynamical System is :

TYPE 1 – characterized by an infinite range of predictability

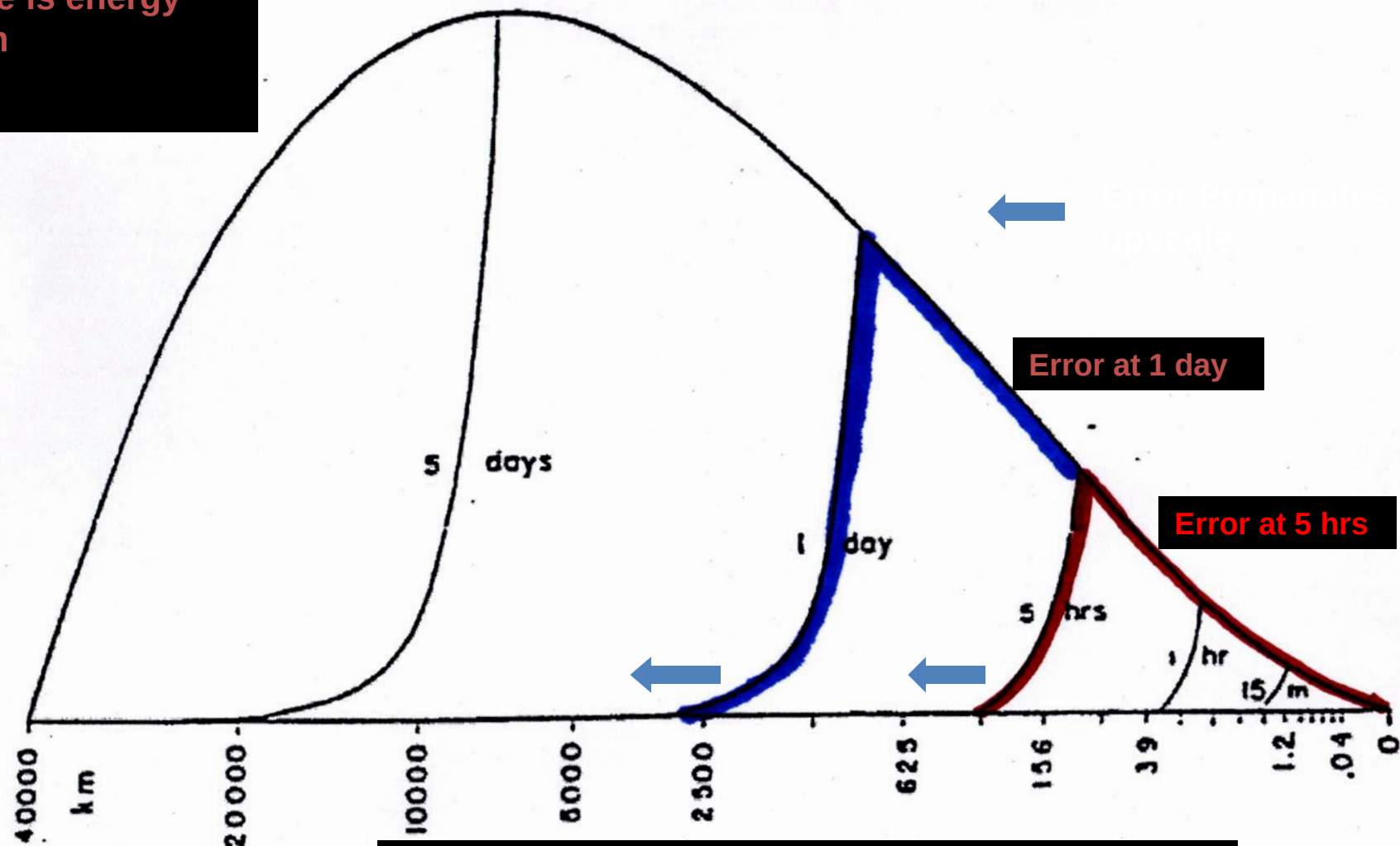
TYPE 2 – the range of predictability is *finite*, but can be *increased indefinitely* by decreasing the size of the initial error

TYPE 3 – the range of predictability is *finite and intrinsically limited*

Does the Weather Constitute a Type 2 or a Type 3 System?



Top curve is energy spectrum
 $(E / \lambda)^{1/4}$



$\lambda^{1/4} = (2 \pi / k)^{1/4}$ (λ = wavelength)



The Growth of Very Small Errors

- Basic Idea – Reduce the Size of the Initial Error by putting it on smaller and smaller scales
- Ultimate Predictability controlled by the predictability time T = time necessary for the error to propagate “upscale” from very, very small initial scale to a finite, pre-chosen scale
- How does T behave as the initial error gets infinitely small? This tells us if we have TYPE 2 or TYPE 3 behavior!
- For a Spectrum $E(k) \sim k^{-3}$ or steeper :
 T becomes infinite (thus TYPE 2)
- For a Spectrum $E(k)$ less steep than k^{-3} :
 T is finite (thus TYPE 3)



Commentary

- ❖ The question remains open whether the observed spectrum with slope $-5/3$ and the underlying arguments are relevant for predictability of the real atmosphere
 - The nature of mesoscales is dominated by moisture-driven circulation and gravity waves
 - The energy propagation in the mesoscales (from observations) is from the large to the small scales, not the other way around
- ❖ In fact, despite 40 years of research, we still cannot definitively state whether the range of predictability cannot be increased indefinitely



Concepts of a Monsoon Model from the Perspective of deterministic forecasting

A Monsoon model must explain

- a. Cross-Equatorial Flow
- b. High Wind Shear
- c. Deep Vertical Heating
- d. Establishment of High/Low pressure Cells and Circulation patterns



A Gill-Type Model:

Gill, A.E., (1980)

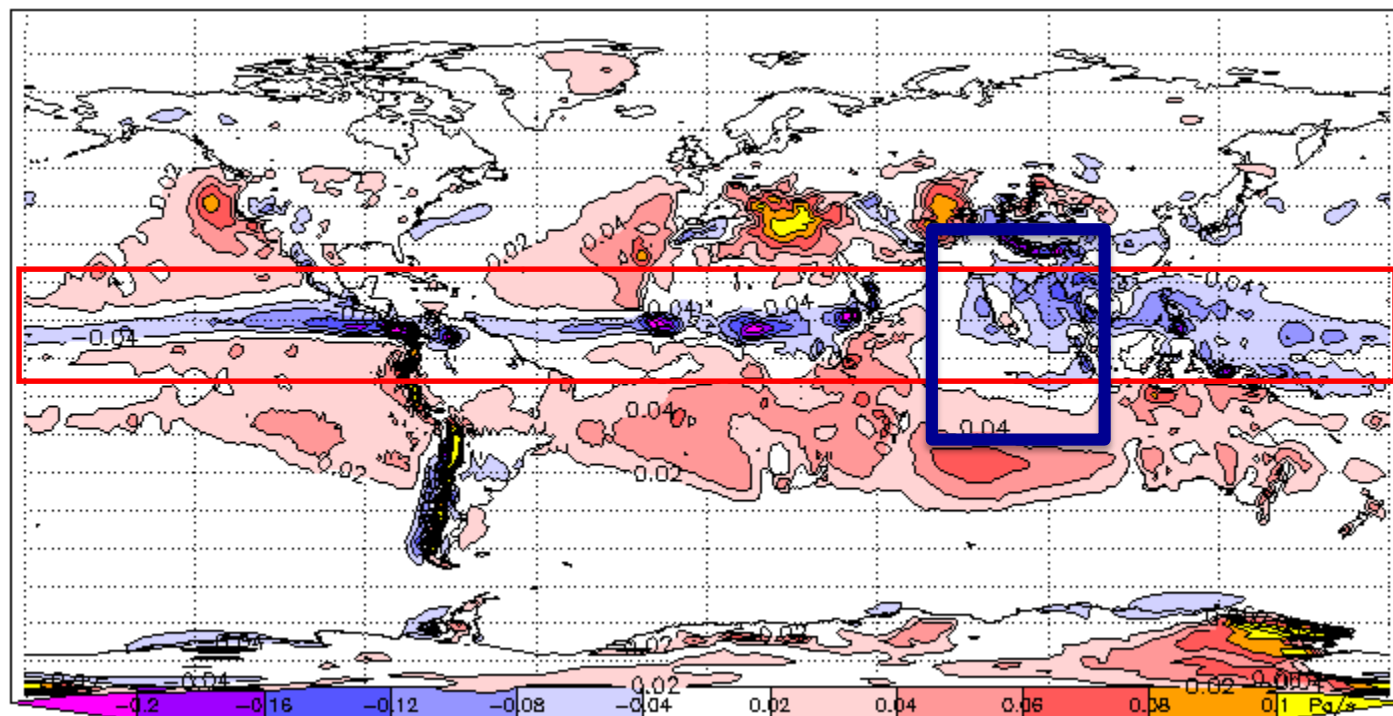
“Some simple solutions for heat-induced tropical circulation.”

Q. J. Royal. Met. Soc., 106,447-462.

**Zhang, Z and T.N. Krishnamurti, JAS, 1993 applied
it to explain monsoon**



Strong Convective heating drives Monsoon Circulation. So a forced circulation Induced by latent heating is the simplest assumption to use the Gill Model.



Ascent movement concentrated along the equator

Vertical velocity at 500 hPa, July average.



The model

- ❖ Heating rate is assumed small enough for linear theory to apply.
- ❖ Heating is equivalent to increasing the amount of high potential temperature fluid (transferring mass from the lower layer to the upper layer).
- ❖ The problem is reduced to solve the shallow-water equations in the tropics. Beta-plane approximation is used (because the motion is confined to the tropics: Coriolis parameter is approximated to β times distance northwards from the equator).



The Basic Heating model

- ❖ Equations are written in a non-dimensional form, where the length scale is the Rossby radius of deformation, so

$$\frac{\partial u}{\partial t} - \frac{1}{2}yv = -\frac{\partial p}{\partial x},$$

$$\frac{\partial v}{\partial t} + \frac{1}{2}yu = -\frac{\partial p}{\partial y},$$

$$\frac{\partial p}{\partial t} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -Q.$$

Momentum eqns.

Continuity eqn.

$$w = \frac{\partial p}{\partial t} + Q,$$

Vertical velocity

- ❖ Q is proportional to the heating rate, the signs being such that if Q is positive (positive heating) the signs of u , v , and p correspond to those at the surface.



The model

❖ Two special cases for the solutions:

- *Symmetric about the equator*

$$Q(x,y) = F(x)D_0(y) = F(x)\exp(-\tfrac{1}{4}y^2).$$

- *Antisymmetric about the equator*

$$Q(x,y) = F(x)D_1(y) = F(x)y\exp(-\tfrac{1}{4}y^2).$$

❖ Response only involves parabolic cylinder functions up to order 3

$$D_0, D_1, D_2, D_3 = (1, y, y^2 - 1, y^3 - 3y)\exp(-\tfrac{1}{4}y^2).$$

Forcing assumed in the neighborhood of $x=0$ and with this form.

$$F(x) = \begin{cases} \cos kx & |x| < L \\ 0 & |x| > L \end{cases}$$

$$k = \pi/2L.$$



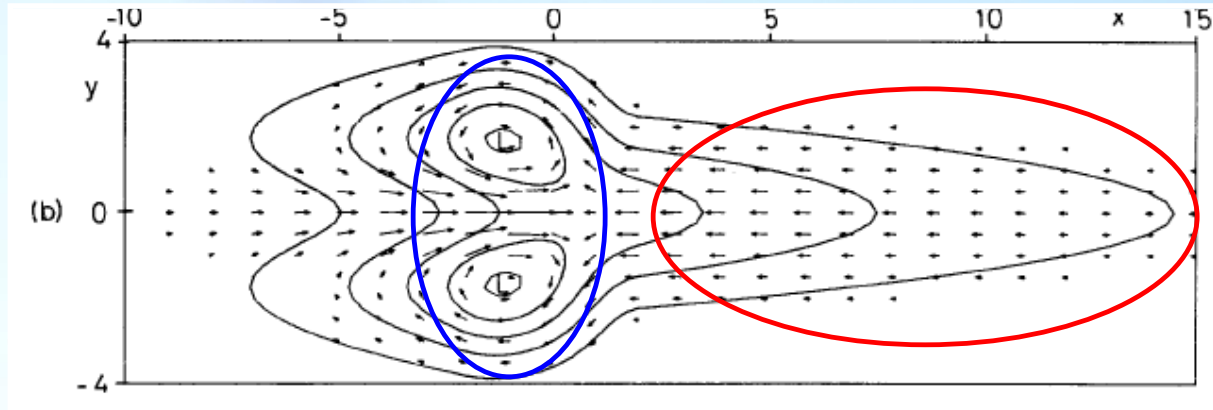
Symmetric forcing: $Q_0 = F(x)$

❖ Two parts of the response:

- The first part ($n=0$) only involves q_0 . Represents a Kelvin wave which is damped out as it progresses eastwards. It moves at a unit speed and decay rate ε . No information carried westwards $\Rightarrow$ solution is zero for $x < -L$.
- If forcing region represents Indonesia area $\Rightarrow$ solution represents the Walker circulation over the Pacific.
- The second part ($n=1$) corresponds to a long planetary wave which propagates westward at speed $1/3$ (spatial decay 3ε). No information carried eastward.



Symmetric forcing: $Q_0 = F(x)$



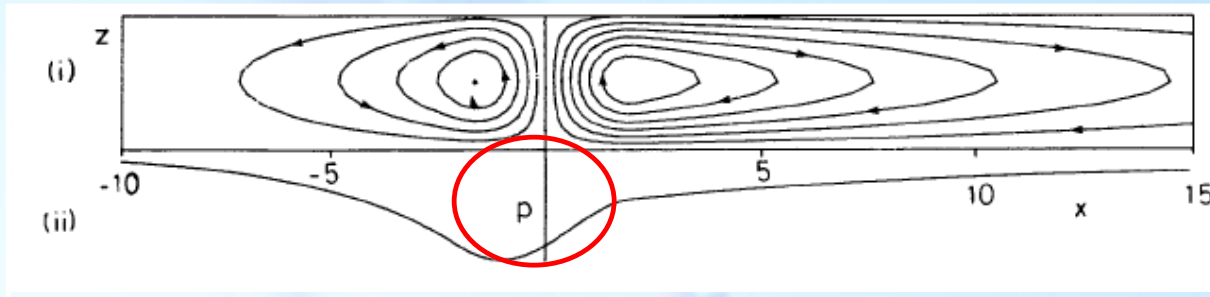
A trough at the equator in the easterly regime to the east of the heating region due to the fact that the winds are in geostrophic balance.

Two cyclones on the north-west and south-west flanks of the heating region: return flow from the poles is situated farther west because of planetary wave, which propagates in that way.

Rising motion in heating region (trough) and subsidence elsewhere. The Walker cell (right-hand) is five times that the Hadley cells show here.

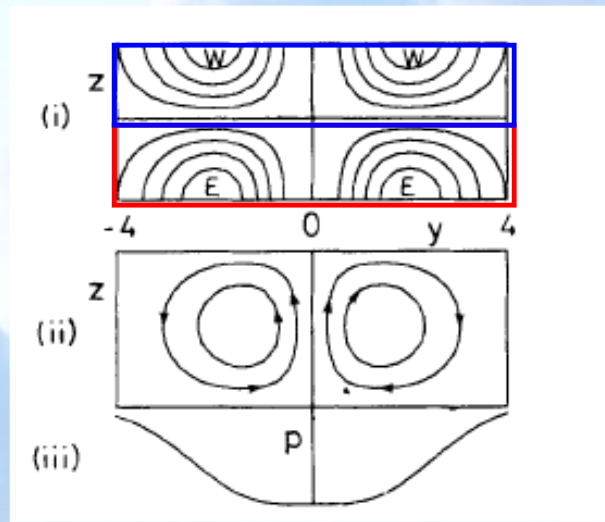


Symmetric forcing: $Q_0 = F(x)$



p'

Meridionally integrated flow

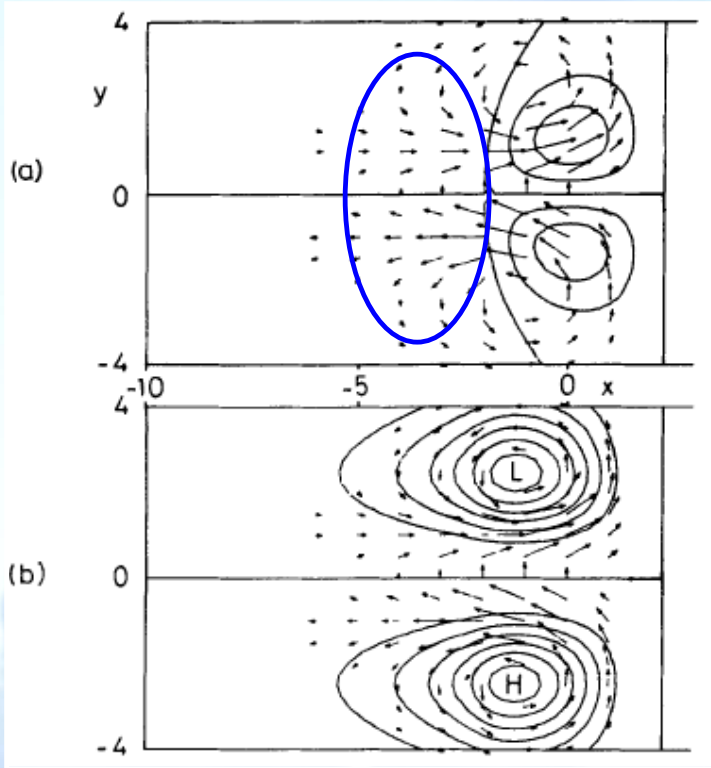


Easterlies in the surface,
westerlies in the upper level

Zonally integrated flow



Antisymmetric forcing: $Q_1 = F(x)$

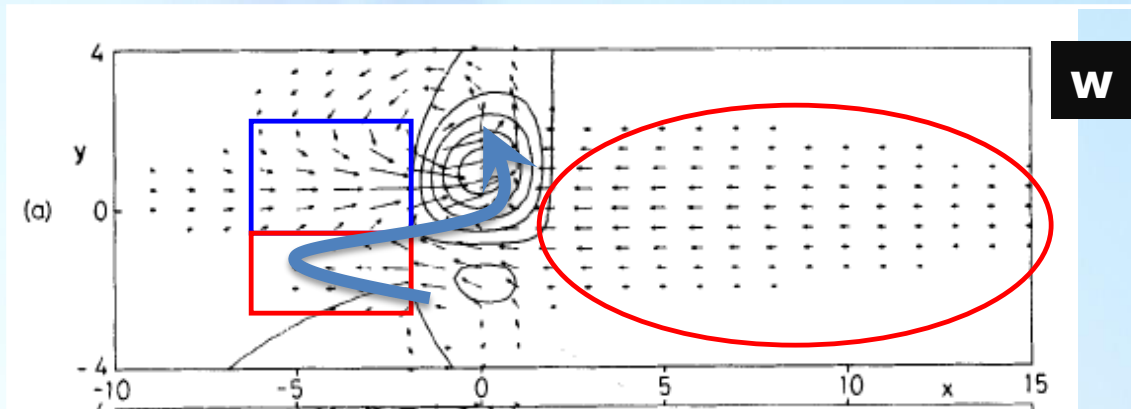


The field is dominated by the motion in the heating region with the same shape than the heating function (positive in NH). Outside the forcing region, subsidence in NH and weak upward motion in SH. No motion for $x > 2$.

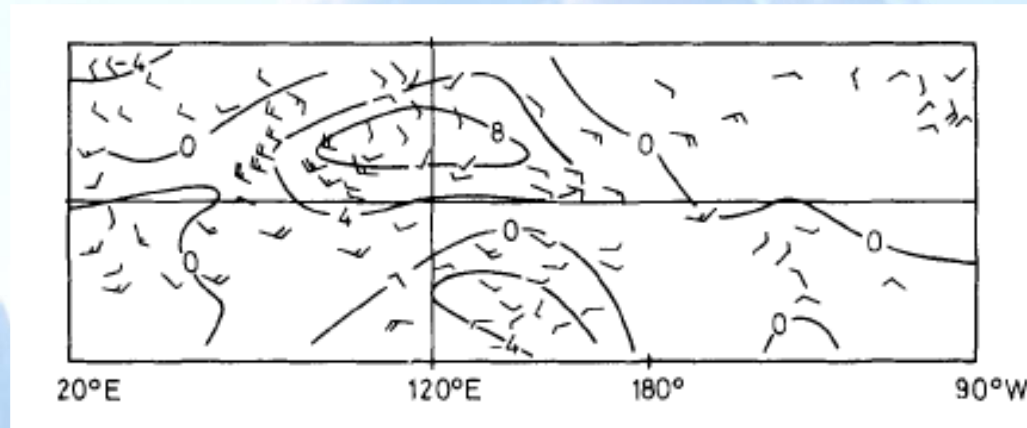
Westerlies to the west but with limited extension (spatial decay for the long $n=2$ planetary wave is 5ε).

Positive p' in the south (where there is an anticyclone) and negative north of the equator (where there is a cyclone). The flow moves down the pressure gradient where it crosses the equator.

The Monsoon type Circulation



$$Q_0 + Q_1$$



w at 500 hPa for June-August season.

Good agreement if the centre of the forcing region is taken 120°E.

Flow to the east of the forcing region is the same than for Q_0 (asymmetric part does not contribute). West of the forcing, the westerly inflow is concentrated between the equator and $y=2$. An easterly flow is found south of the equator.

Atmospheric Modeling Hindcasts

❖ TOGA MONEG simulations

- 1987 and 1988
- A large fraction of the variability of the simulated Indian monsoon was forced by the SST variations over the Pacific (Palmer et al. 1992)

❖ AMIP simulations

- January 1979- December 1988
- SST and Sea ice from observations as boundary conditions
- Sperber and Palmer (1996)
- Gadgil and Sajini (1998)
- Wang et al (2004)

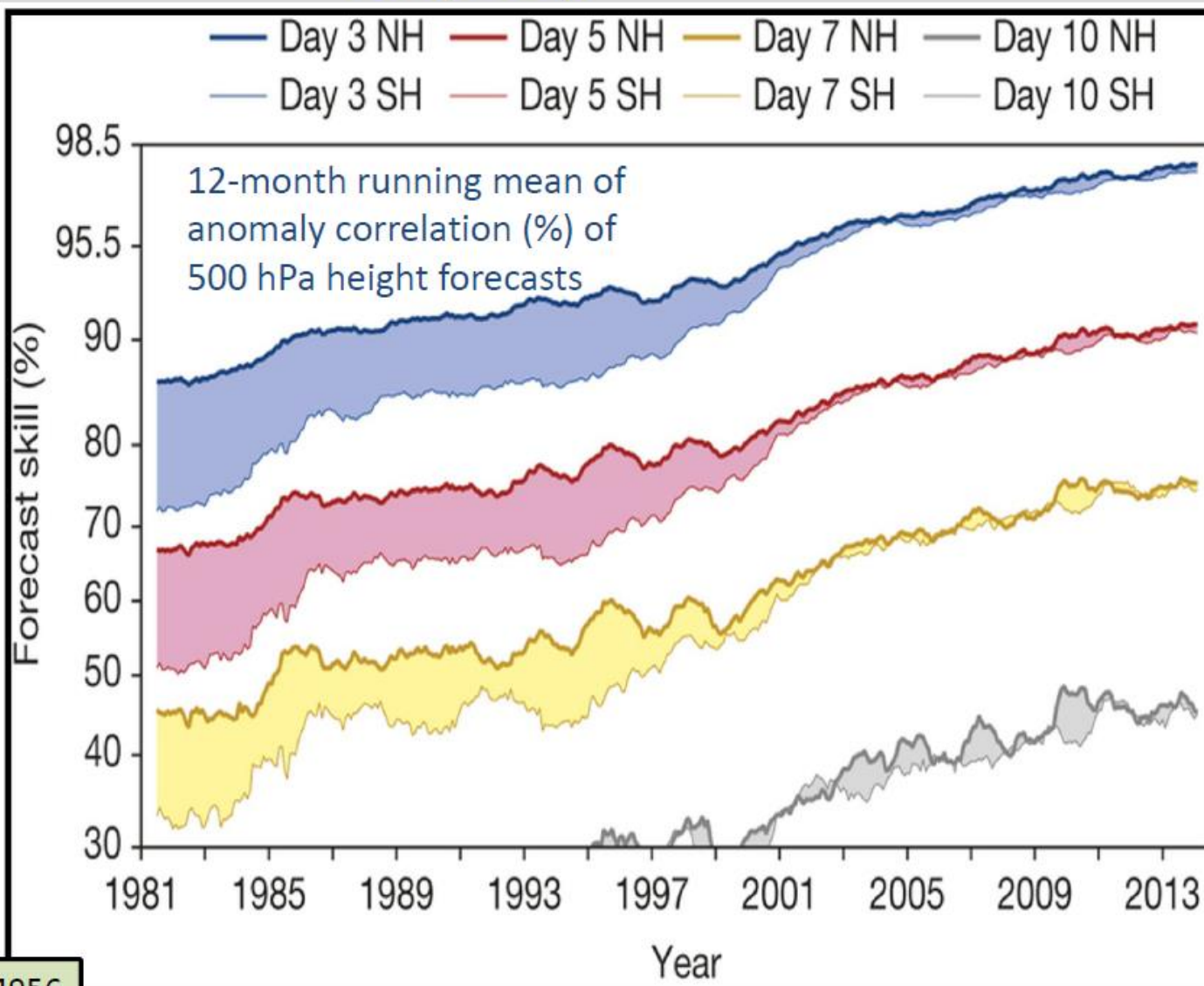


General Inferences from AMIP results

- ❖ Two classes of models (GS 1998)
 - Models well simulated the seasonal migration of rain-belt over the Indian region (class-1)
 - Models not well simulated (class-2)
- ❖ mean monsoon rainfall and the inter-annual variability of Indian summer monsoon interrelated.
- ❖ AGCMs, when forced by observed sea surface temperature (SST), had problems in predicting inter-annual variation of Indian monsoon



Improvement in forecast skill (better physics, observation, computer)



Improvement of 1 day per decade

**Therefore, Can Ocean Atmosphere coupling
generate the monsoonal flow?**

Or

Does coupled model does a better job?

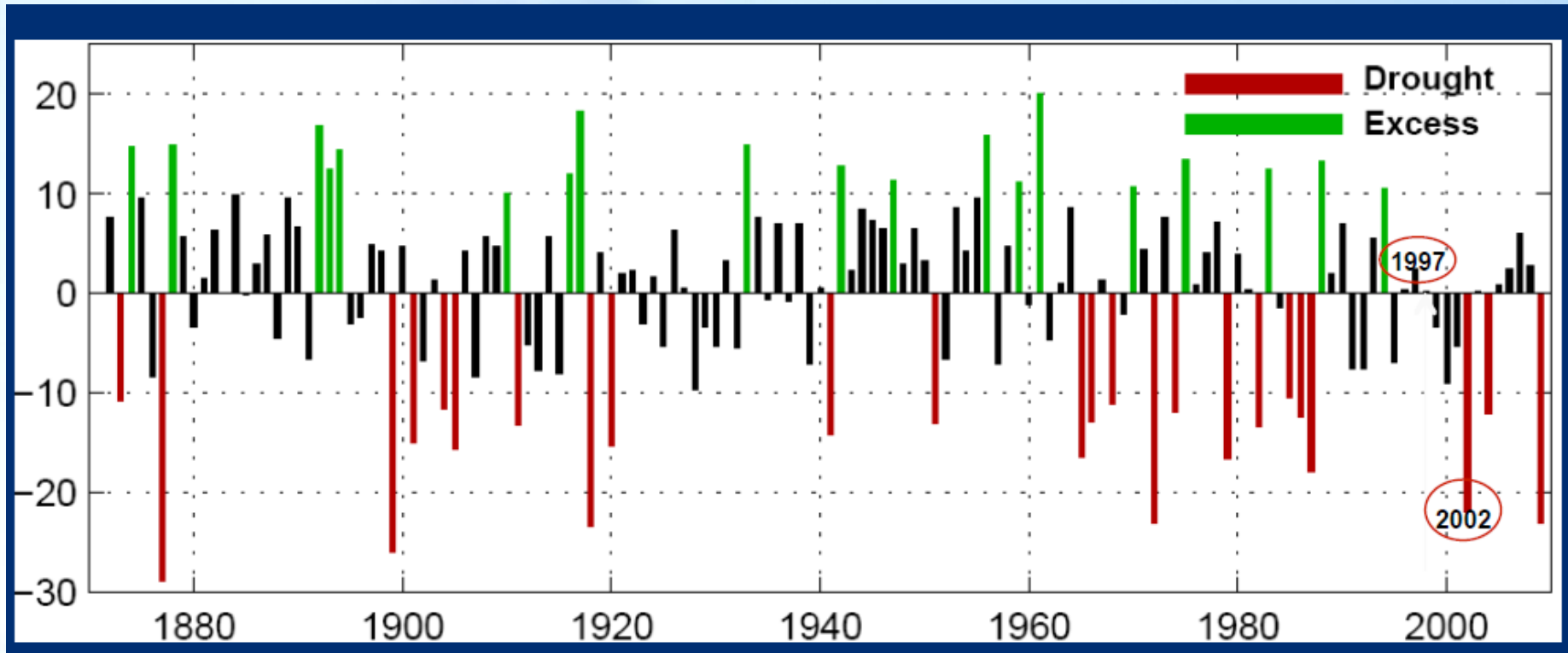


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The ENSO Monsoon Relationship



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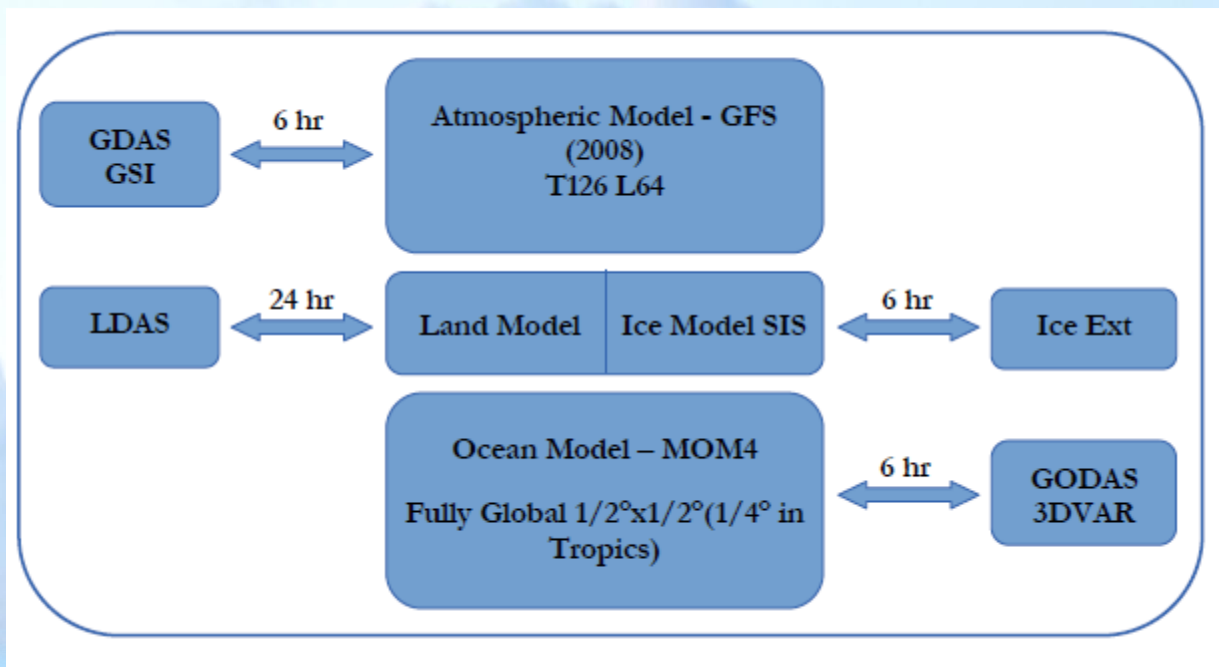
Air –Sea Interaction and subseasonal monsoon

Ravichandran, 2001). Using the observations from special field experiment like BOBMEX and JASMINE, Bhat et al., (2001), and Webster et al., (2002) have shown large intraseasonal fluctuation of the net heat flux. Using the high resolution TMI (TRMM microwave imager) data Sengupta et al., (2001) showed that the intraseasonal oscillation of SST has a large horizontal scale comparable to that of its atmospheric counterpart and maximum amplitude of fluctuation ranges between 0.6° - 0.8°C . It shows coherent northward propagation along with OLR (convection), surface wind and net heat flux during the summer monsoon of 1998-2000. Warm (cold) SST follow the rainfall break (active) phases over Indian region with a time lag of about 7-10 days. Vecchi and Harrison (2002) suggested that such phase relationship may be exploited to predict the active break phases of ISO over the Indian region. The net heat flux shows a fluctuation in amplitude of around $70\text{-}90\text{ Wm}^{-2}$ and more importantly leads the northward propagation of SST by about 1 week in the Bay of Bengal (Sengupta et al., 2001). Such a quadrature relationship suggests that oceanic ISO is driven by atmospheric ISO through net heat flux in the Bay of Bengal. However such quadrature relationship is not found in other regions, e.g. equatorial Indian Ocean and west coastal region in the Arabian Sea



Coupled Modelling Framework for monsoon forecast

Under National Monsoon Mission the operational version of coupled model forecast is CFSv2 from NCEP



CFSv2

The NCEP Climate Forecast System version 2 (CFSv2; Saha et al. 2013) is a state-of-the-art coupled climate model developed by the National Center for Environmental Prediction (NCEP), USA.

It has been implemented to the Prithvi High Performance Computer (HPC) at Indian Institute of Tropical Meteorology (IITM), Pune with support from the Monsoon Mission Project of Ministry of Earth Science, Govt. of India.

It is a fully coupled ocean- atmosphere-land model with advanced physics, increased resolution and refined initialization as compared to earlier version (CFSv1).



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Model Components

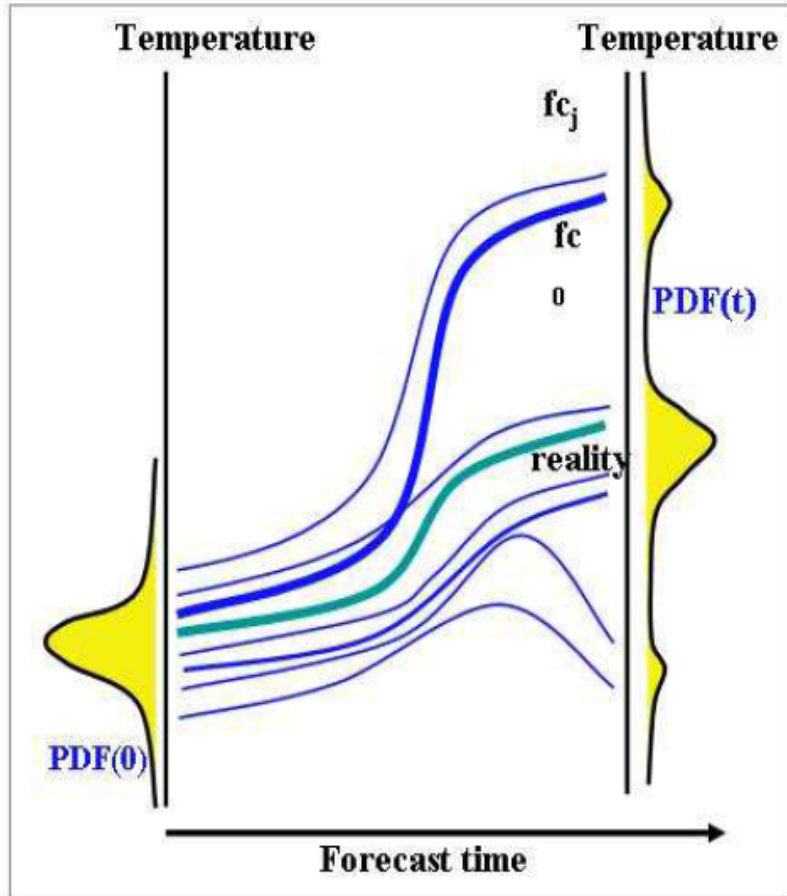
The atmospheric component of CFSv2 is Global Forecast System (GFS) with spectral resolution of T126 and 64 hybrid vertical levels (Moorthi et al. 2001; Yang et al. 2006)

The ocean component is Geophysical Fluid Dynamics Laboratory (GFDL) Flexible Modeling System (FMS) Modular Ocean Model version 4p0d (MOM4; Griffies et al. 2004).

The convective parameterization utilizes simplified Arakawa-Schubert convection with momentum mixing. The model also implements orographic gravity wave drag based on Kim and Arakawa (1995) approach and sub-grid scale mountain blocking (Lott and Miller 1997).



Types of forecasts



Deterministic model

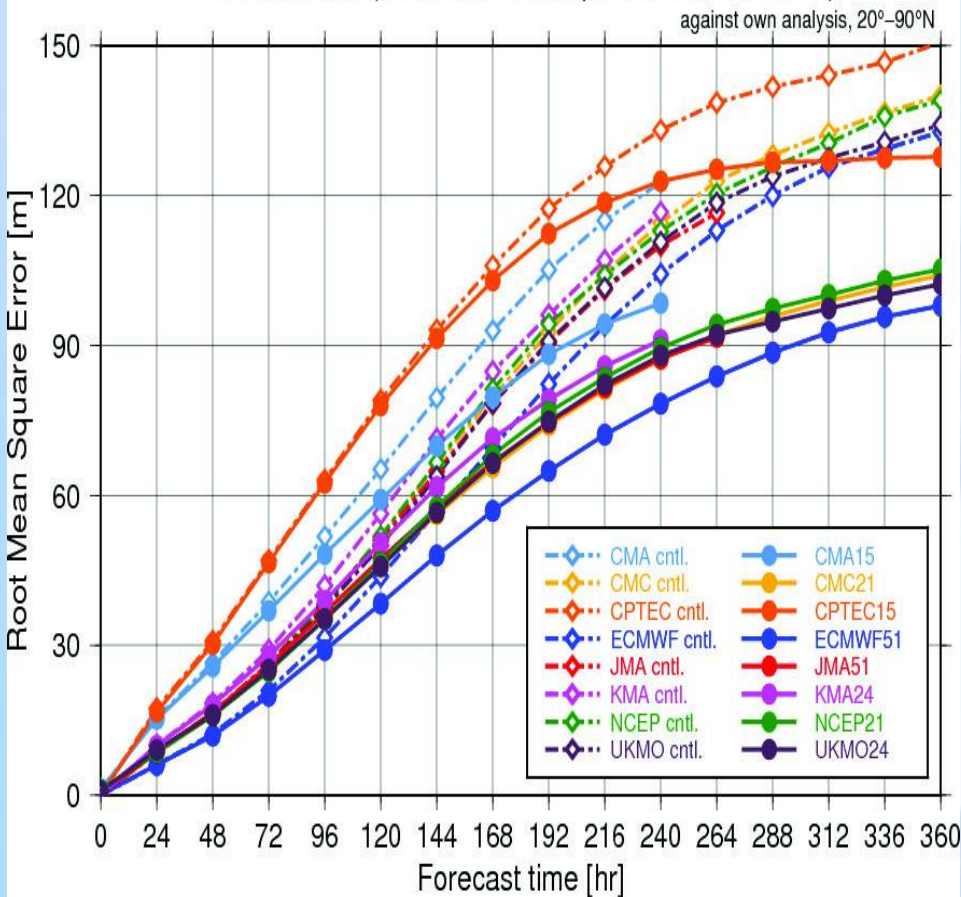
- Decide for in initial state and start a single integration

Ensemble prediction

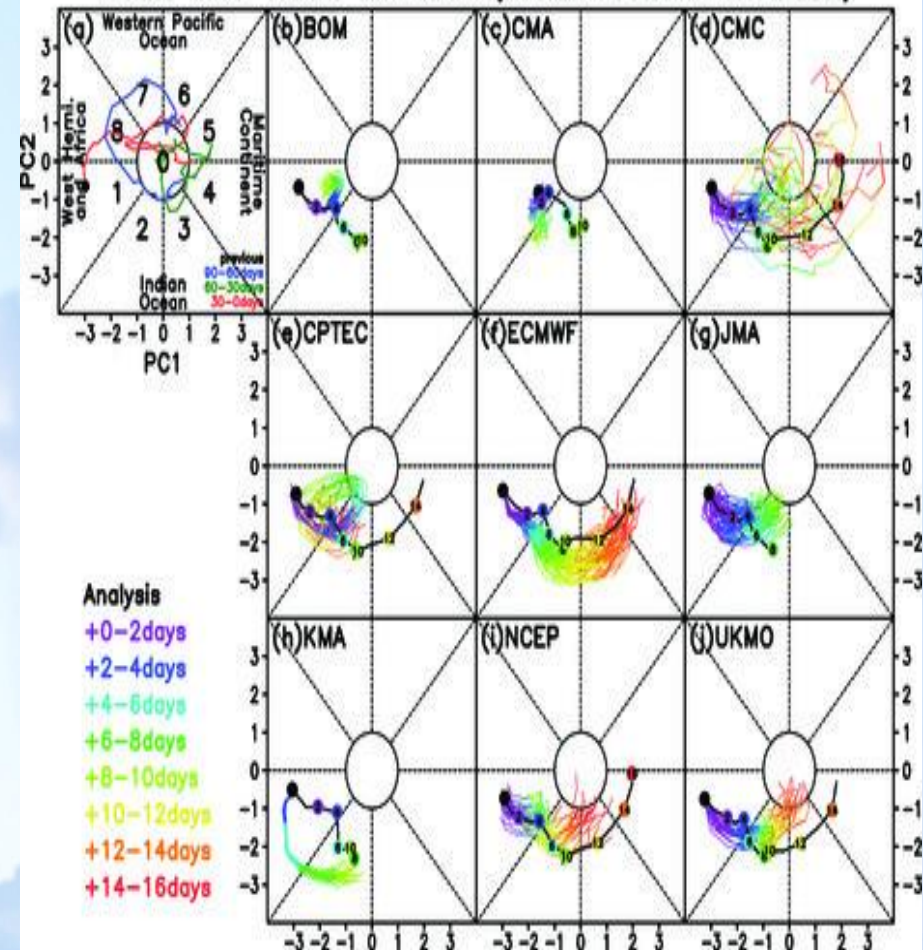
- Make many integrations (e.g. from different initial states, or multi model or analogue ensemble...)

ENSEMBLE FORECASTING

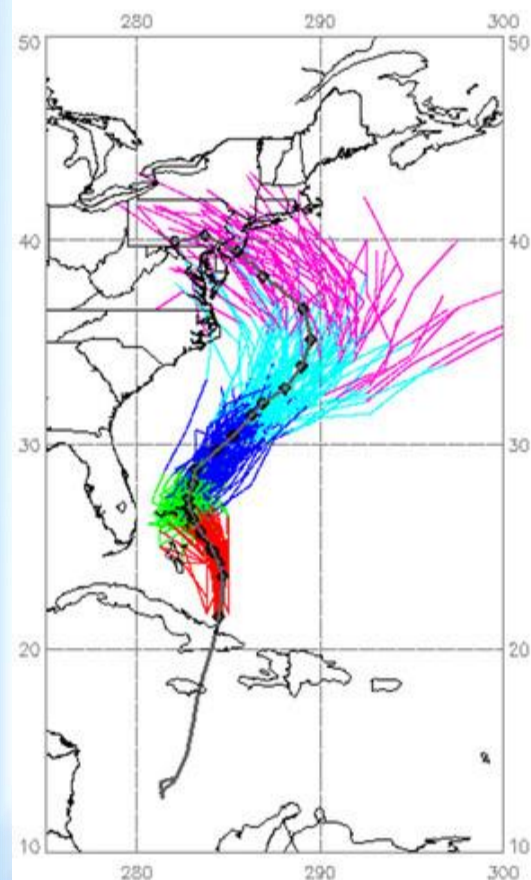
TIGGE medium-range ensemble forecasts
Z500 RMSE (Northern Hemisphere, DJF2013/14)



TIGGE MJO index forecast (Initial: 2009040112UTC)



a) Ensemble forecast tracks



— T0-24 — T72-96
— T24-48 — T96-120
— T48-72

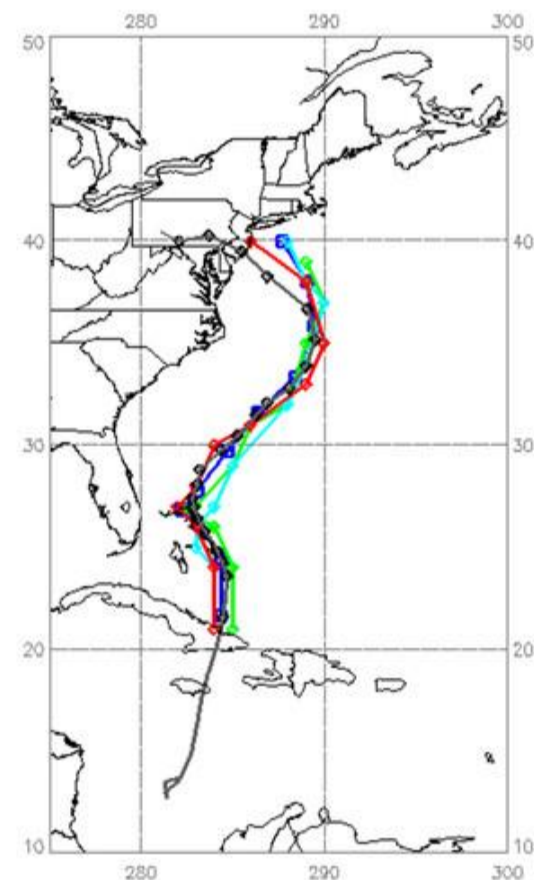
b) Strike probabilities



Probability the storm will pass within 75 miles

5-19% 20-39% 40-59% 60-79% 80-100%

c) Ensemble mean forecast tracks



—x— ncep —ukmo
—o— ecmwf —o— ukmo_ecmwf_ncep
Symbols plotted every 12 hours

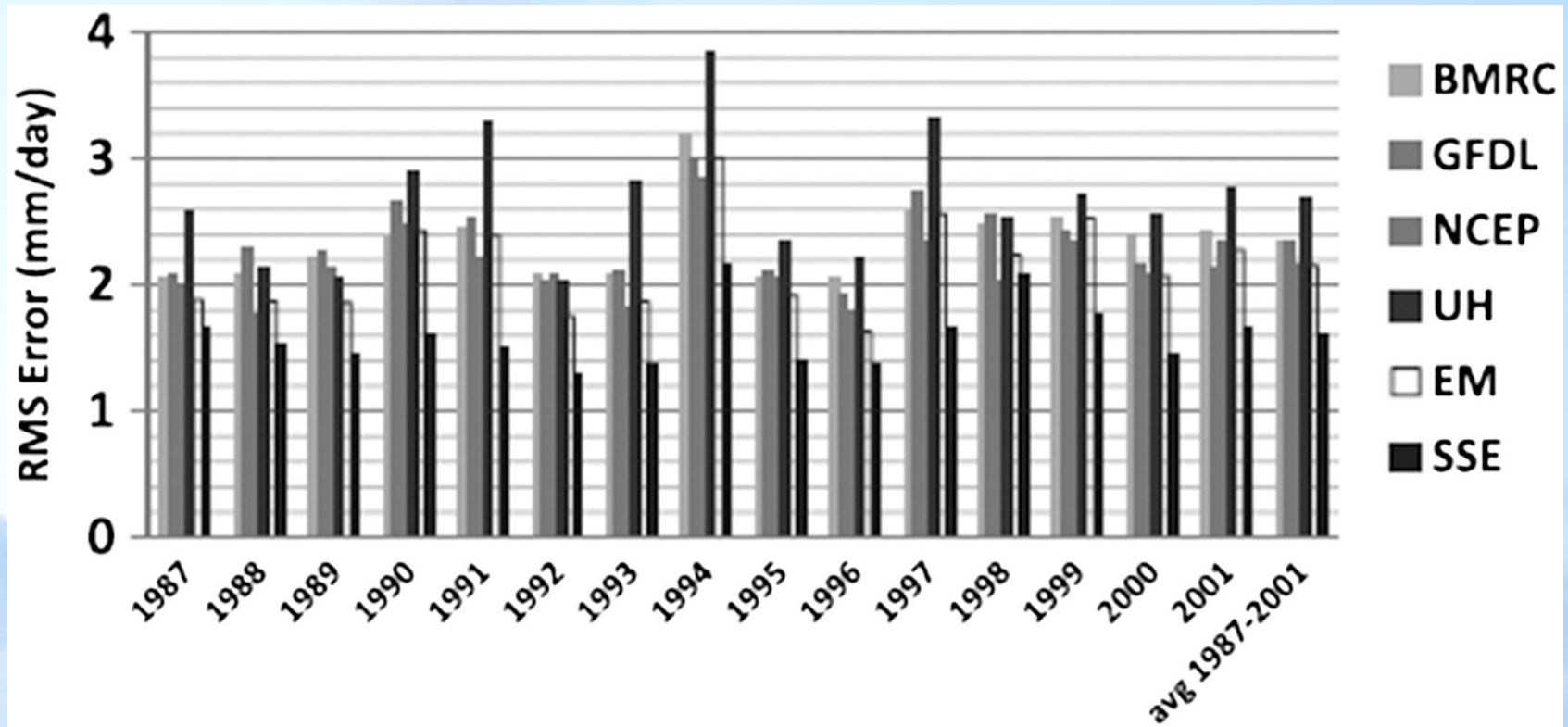


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Super Ensembles: Benefits of Multi Model Super Ensemble



T N Krishnamuti et al., 2013 Rev. Geophys



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Operational Seasonal Forecasting using Dynamical Models

3.b. Updated Forecast for the 2021 Southwest Monsoon Rainfall over the Country based on the Multi Model Ensemble (MME) Forecasting System

The updated MME forecast for 2021 southwest Monsoon season rainfall has been computed using various coupled global model forecasts with May initial conditions. Climate models with the highest forecast skills over the Indian monsoon region including MMCFS have been used to prepare the MME forecast.

The updated MME forecast also suggests that the monsoon rainfall during the 2021 monsoon season (June to September) averaged over the country as a whole is likely to be normal (96-104% of LPA).

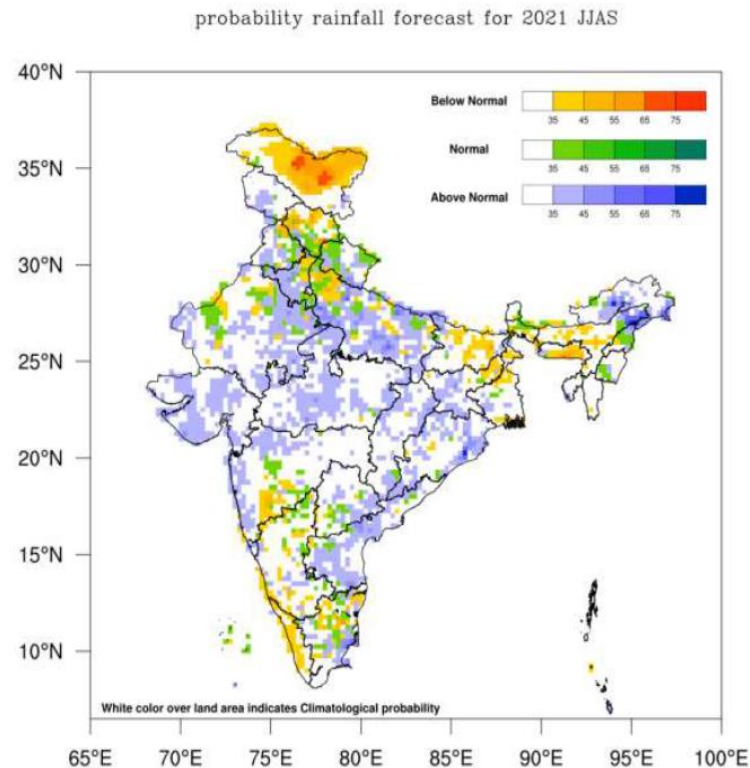


Fig.1. Updated Probability forecast of tercile categories* (below normal, normal and above normal) for the seasonal rainfall over India during the 2021 southwest monsoon season (June - September). The figure illustrates the most likely categories as well as their probabilities. The white shaded areas within the land area represent climatological probabilities. The probabilities were derived using the MME forecast prepared from a group of coupled climate models. (*Tercile categories have equal climatological probabilities, of 33.33% each).

Major Model Biases observed in the CFS model

- ❖ Dry bias over Indian landmass
- ❖ Cold and dry bias in troposphere
- ❖ Cool SST bias in the tropics
- ❖ Deep Snow over Eurasia and late melting of Snow over Eurasia



Work on model Improvements

❖ Parameterization

❖ Critical relative humidity sensitivity expts.

❖ High resolution model experiments



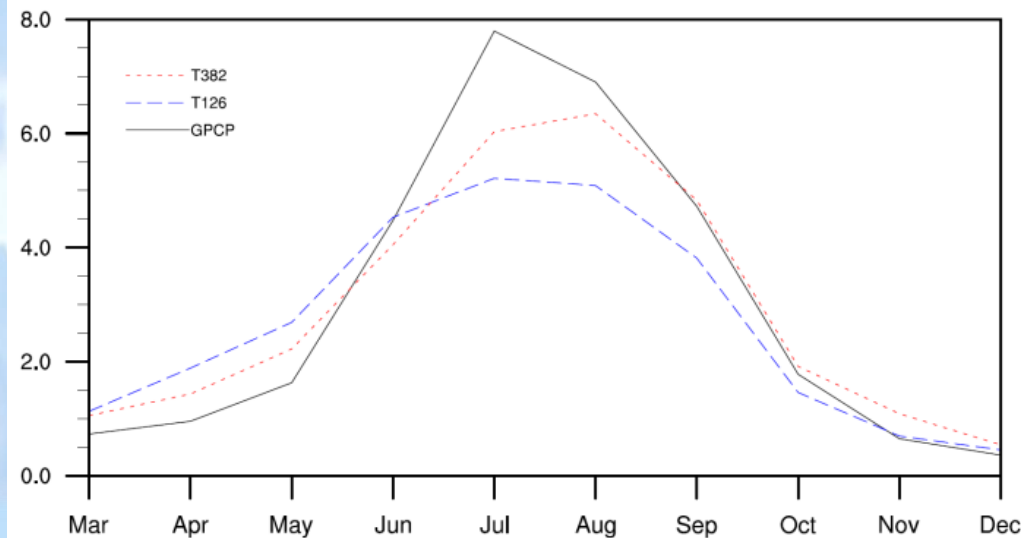
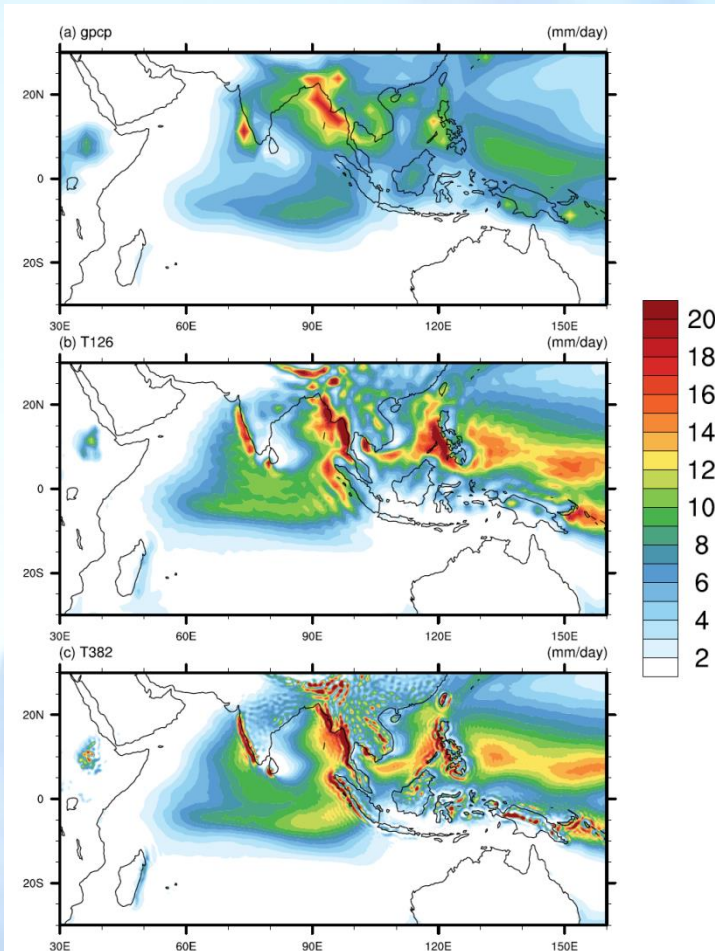
High resolution Seasonal Prediction Experiments

CFSV1	CFSV2	CFSV2/IITM
T62 horizontal resolution (~200 Km)	T126 horizontal resolution (~100 Km)	T382 horizontal resolution (~38Km)
Sigma vertical coordinate with 28 levels with top pressure ~3 hPa	Sigma-pressure hybrid vertical coordinate with 64 levels with top pressure ~0.266 hPa	
Simplified Arakawa-Schubert convection	Simplified Arakawa-Schubert convection with momentum mixing	
Tiedtke (1983) shallow convection	Tiedtke (1983) shallow convection modified to have zero diffusion above the low level inversions	
Orographic gravity wave drag based on GLAS/GFDL approach	Orographic gravity wave drag based on Kim and Arakawa(1995) approach and sub-grid scale mountain blocking following Lott and Miller (1997)	

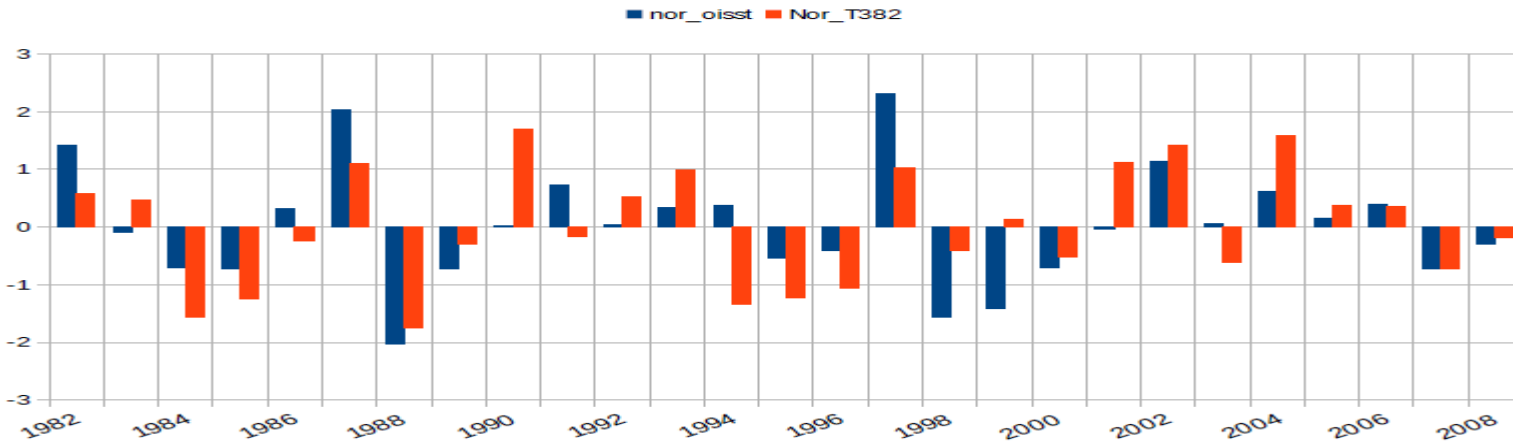


High Resolution Seasonal Prediction Expt.

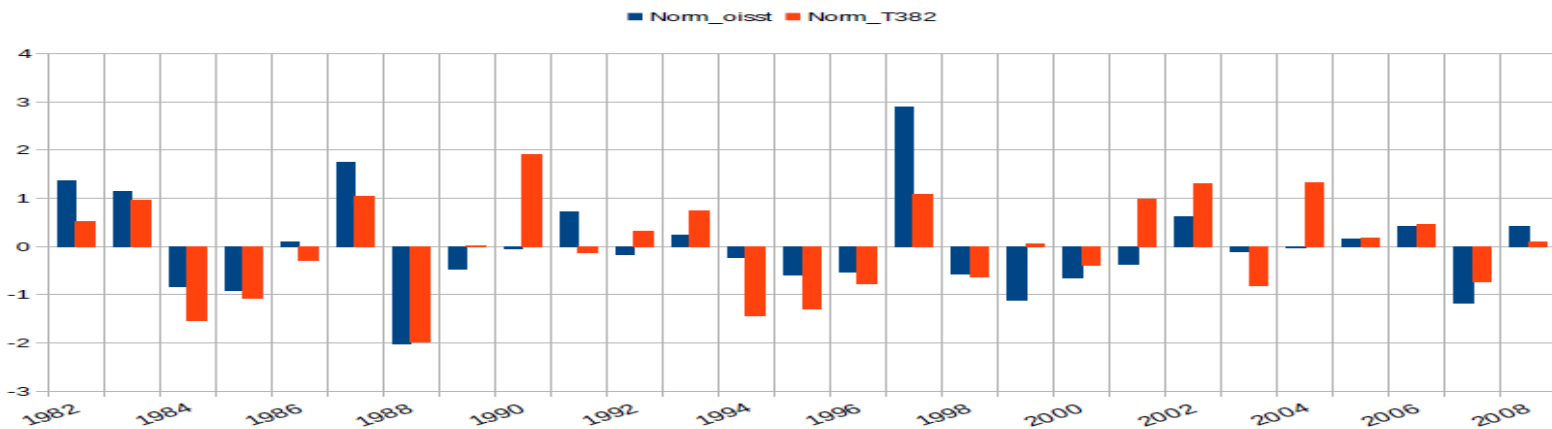
Improvement in Land Rainfall



T382L64 Skill of SST indices in JJAS



Nino 3.4
C.C=0.62

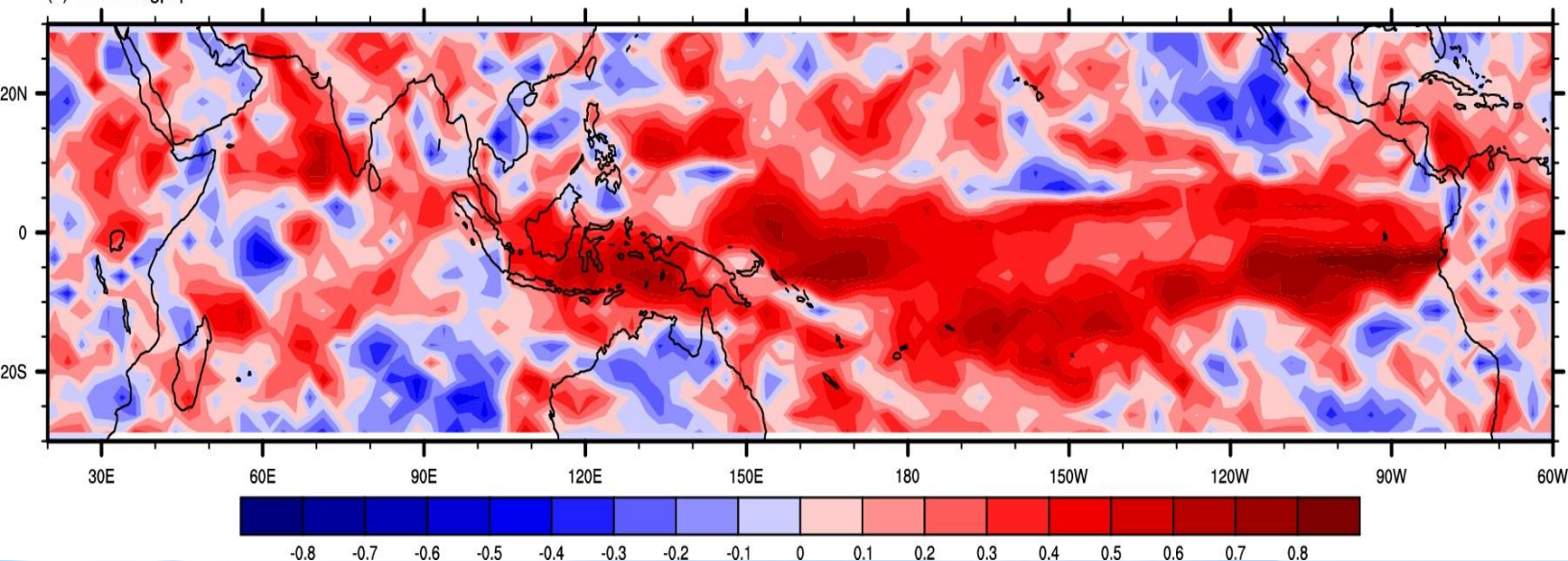


Nino 3
C.C=0.64



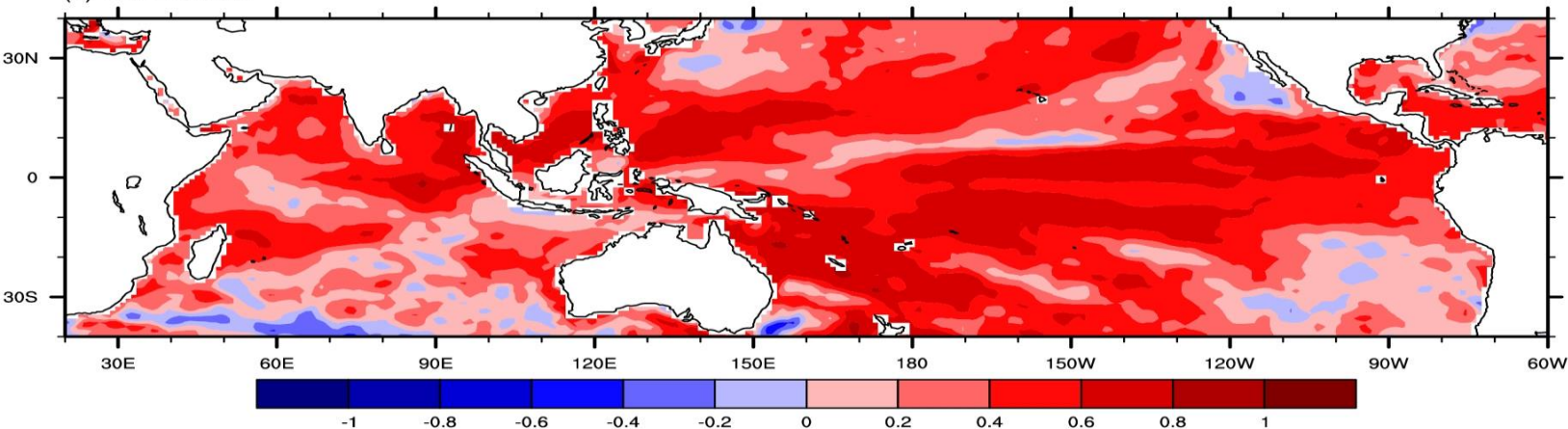
T382L64 Skill of Rainfall/SST

(a) T382. vs.gpcp



GPCP VS
T382

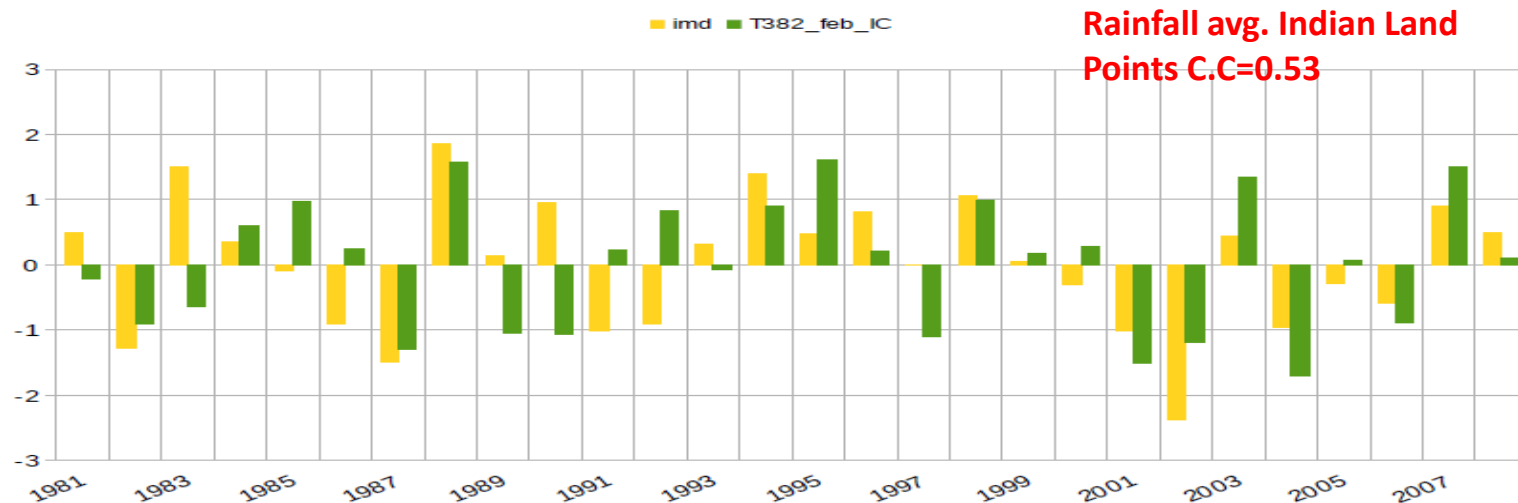
(a) T382. vs.oisst



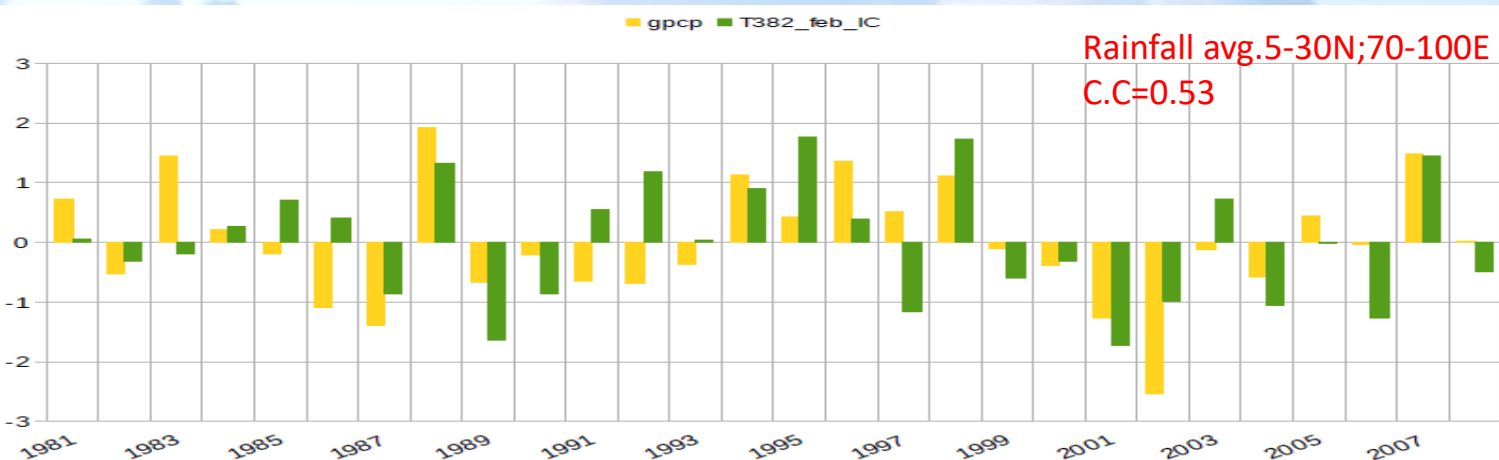
ERSST VS
T382



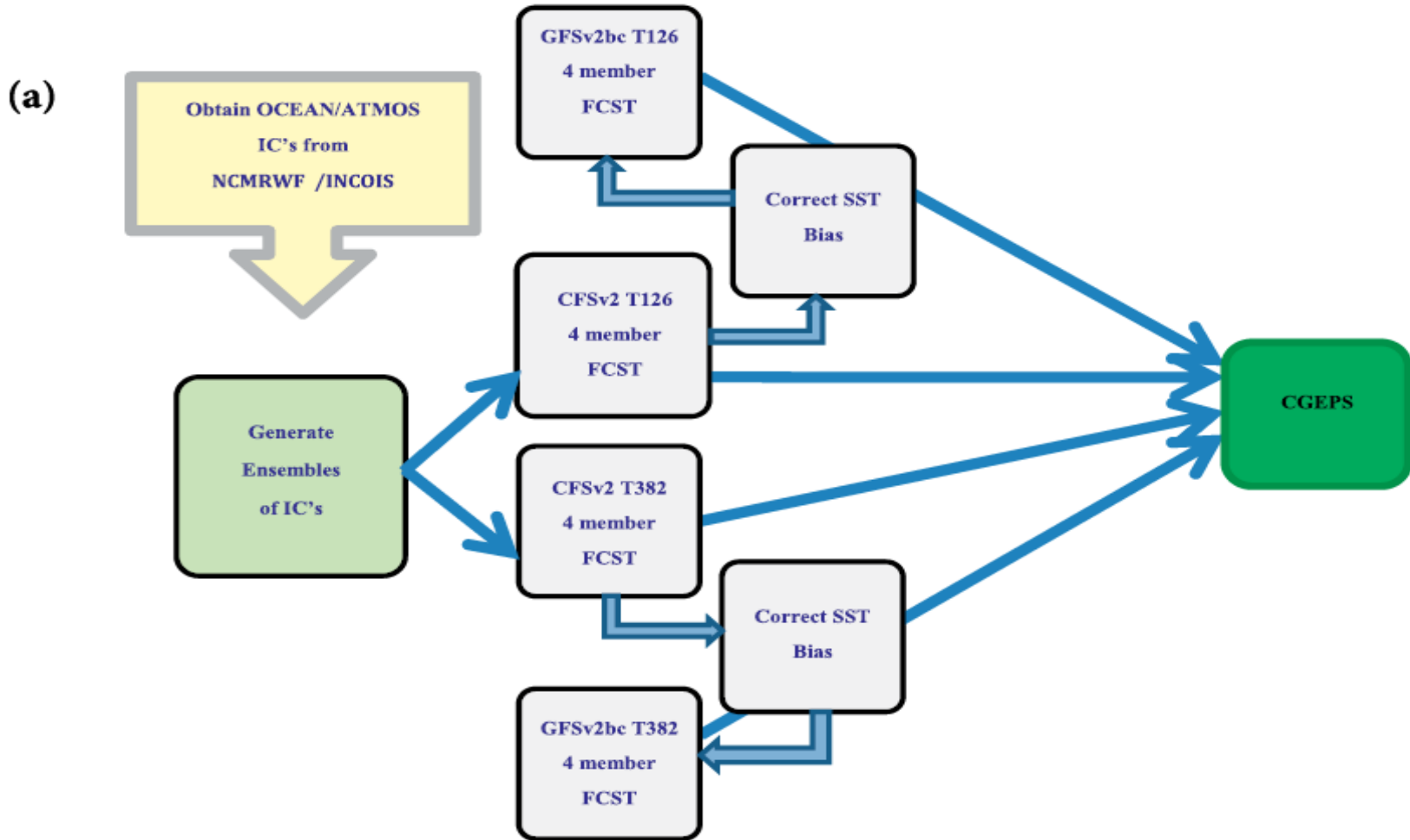
T382L64 Skill of Rainfall



Interannual variability has increased compared to low resolution Model, very close to observations



Multi- Model ENSEMBLE strategy using CFS: extended range forecast



**In the next lecture operational
extended range forecast would be
discussed in detail**

Thank You



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